WORKED SOLUTION

STEP I 2005, Question 7

Pure • Telescoping Products

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q7

The key idea

All three products *telescope*: write each factor as a ratio $g(r+1)/g(r)$, or as a product of such ratios, and almost everything cancels, leaving only the first and last values. Parts (i) and (ii) are warm-ups that show the mechanism. In part (iii) the work is to spot that the trigonometric factor is secretly such a ratio. Putting it over the common denominator $\sin\frac{(2r-1)\pi}{n}$ turns the numerator into a compound-angle sine.

Solution

Part (i)

Writing the product out,

$$\prod_{r=1}^{n}\frac{r+1}{r}=\frac{2}{1}\times\frac{3}{2}\times\frac{4}{3}\times\cdots\times\frac{n+1}{n}.$$

Each numerator cancels the denominator of the next factor, so only the last numerator and the first denominator survive:

$$\prod_{r=1}^{n}\frac{r+1}{r}=\frac{n+1}{1}=n+1.$$

Part (ii)

Factorise the numerator, $r^2-1=(r-1)(r+1)$, and split each factor into two:

$$\frac{r^2-1}{r^2}=\frac{r-1}{r}\times\frac{r+1}{r}.$$

Since the product of products is the product of the factors, we can treat the two families separately (here $n \geq 2$):

$$\prod_{r=2}^{n} \frac{r-1}{r} = \frac{1}{2} \times \frac{2}{3} \times \cdots \times \frac{n-1}{n} = \frac{1}{n},$$
$$\prod_{r=2}^{n} \frac{r+1}{r} = \frac{3}{2} \times \frac{4}{3} \times \cdots \times \frac{n+1}{n} = \frac{n+1}{2}.$$

Both telescope exactly as in part (i). Multiplying,

$$\prod_{r=2}^{n} \frac{r^2-1}{r^2} = \frac{1}{n} \times \frac{n+1}{2} = \frac{n+1}{2n}.$$

Part (iii)

Write $\alpha = \dfrac{2\pi}{n}$ and $\beta_r = \dfrac{(2r-1)\pi}{n}$, so the rth factor is $\cos\alpha + \sin\alpha \cot\beta_r$.

Ask yourself How can $\cos\alpha + \sin\alpha \cot\beta$ be a ratio of two similar things?

Write $\cot\beta = \cos\beta / \sin\beta$ and use a common denominator. The numerator is then the expansion of a compound angle:

$$\cos\alpha + \sin\alpha \frac{\cos\beta}{\sin\beta} = \frac{\sin\beta\cos\alpha + \cos\beta\sin\alpha}{\sin\beta} = \frac{\sin(\alpha+\beta)}{\sin\beta}.$$

This needs $\sin\beta_r \neq 0$, and it is here that we use the condition that n is even. We would have $\sin\beta_r = 0$ only if $(2r-1)/n$ were an integer. But $2r-1$ is odd and n is even, so an odd number can't be a multiple of n. So every factor is defined, and every denominator below is non-zero.

Now $\alpha + \beta_r = \dfrac{2\pi + (2r-1)\pi}{n} = \dfrac{(2r+1)\pi}{n} = \beta_{r+1}$, so

$$\cos\alpha + \sin\alpha\cot\beta_r = \frac{\sin\beta_{r+1}}{\sin\beta_r}.$$

This is exactly the shape of part (i), with $g(r) = \sin\beta_r$. The product telescopes:

$$\prod_{r=1}^{n} \frac{\sin\beta_{r+1}}{\sin\beta_r} = \frac{\sin\beta_2}{\sin\beta_1} \times \frac{\sin\beta_3}{\sin\beta_2} \times \cdots \times \frac{\sin\beta_{n+1}}{\sin\beta_n}$$
$$= \frac{\sin\beta_{n+1}}{\sin\beta_1} = \frac{\sin(\frac{(2n+1)\pi}{n})}{\sin\frac{\pi}{n}} = \frac{\sin(2\pi + \frac{\pi}{n})}{\sin\frac{\pi}{n}}.$$

Since sin has period 2π, the numerator equals the denominator, and

$$\prod_{r=1}^{n} \left(\cos\tfrac{2\pi}{n} + \sin\tfrac{2\pi}{n}\cot\tfrac{(2r-1)\pi}{n}\right) = 1.$$

Checks

- Part (i) with $n = 3$: $2 \cdot \frac{3}{2} \cdot \frac{4}{3} = 4 = n + 1$.
- Part (ii) with $n = 2$: the single factor is $\frac{3}{4}$, and $\frac{n+1}{2n} = \frac{3}{4}$. As $n \to \infty$ the product tends to $\frac{1}{2}$, which is a well-known infinite product.
- Part (iii) with $n = 4$: $\cos \frac{\pi}{2} = 0$ and $\sin \frac{\pi}{2} = 1$, so the factors are $\cot \frac{\pi}{4}$, $\cot \frac{3\pi}{4}$, $\cot \frac{5\pi}{4}$ and $\cot \frac{7\pi}{4}$, that is, 1, -1, 1 and -1, with product 1. We also checked numerically that the product equals 1 (to within 10^{-8}) for every even n up to 60.
- If n is odd, the factor with $r = \frac{n+1}{2}$ involves $\cot \pi$, which is undefined. That explains the condition.

Common slips

- In part (ii), starting the cancellation from $r = 1$: the lower limit is 2, so the first factors are $\frac{1}{2}$ and $\frac{3}{2}$.
- In part (iii), stopping at $\sin \frac{(2n+1)\pi}{n} / \sin \frac{\pi}{n}$ without using periodicity to simplify it to 1.
- Not explaining where "n even" is used. It is exactly what makes every cotangent, and every denominator, well defined.

Beyond STEP

Telescoping is the multiplicative cousin of the telescoping sums you meet with partial fractions. Taking logarithms turns one into the other. Part (ii) is a finite version of the infinite product $\prod_{r \geq 2}(1 - 1/r^2) = \frac{1}{2}$. Infinite products like this, and Euler's product

$$\sin \pi x = \pi x \prod_{r \geq 1} \left(1 - \frac{x^2}{r^2}\right),$$

are studied properly in a first analysis course, where convergence of products is defined via the convergence of sums of logarithms. Abbott's *Understanding Analysis* is a good place to build the convergence ideas you'll need.

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